

Program : Diploma in Architecture	
Course Code : 3189	Course Title: CAD Lab for Architecture
Semester : 3	Credits: No Credit
Course Category: Program Core	
Periods per week: 4 (L:0, T:0, P:4)	Periods per semester: 60

Course Objectives:

- To impart the knowledge of CAD commands for drawing 2D building drawings required for various civil engineering applications.
- To generate a detailed project report

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic computer knowledge		Introduction to IT systems	1
Knowledge of Engineering Graphics		Engineering Graphics	1

Course Outcomes:

On completion of the course, the student will be able to:

Con	Description	Duration (Hours)	Cognitive level
CO1	Develop geometric figures using various commands	15	Applying
CO2	Apply preliminary settings of CAD work sheet and develop plan of various buildings	14	Applying
CO3	Develop detailed elevation and section of various type of buildings	14	Applying
CO4	Develop detailed drawings of rain water harvesting tank, septic tank and service plan of the building	14	Applying
	Series Test	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			2
CO2				3			2
CO3				3			2
CO4				3			2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

On completion of the course, the student will be able to:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Develop geometric figures using various commands		
M1.01	Create an electronic worksheet by setting, limits, units etc and save to destination.	5	Applying
M1.02	Draw simple geometric compositions	5	Applying
M1.03	Annotate the drawing by using commands in proper scale and modify them.	5	Applying
CO2	Apply preliminary settings of CAD work sheet and develop plan of various buildings		
M2.01	Develop plan of single storied buildings	12	Applying
M2.02	Apply dimension Styles and hatch pattern to drawings	2	Applying
	Series Test - I	1.5	
CO3	Develop plan, elevation and section of various type of buildings		
M3.01	Develop elevation and sectional views of single storied	8	Applying
M3.02	Detail building components like Doors and Windows by Creating blocks.	6	Applying
CO4	Apply layer and line type properties		
M4.01	Work with layers in drawings	10	Applying
M4.02	Familiarize with properties, groups, utilities and clipboard usage.	4	Applying
	Series Test II	1.5	

Text /Reference:

T/R	Book Title/Author
R1	AutoCAD Essentials, Autodesk official Press, John Wiley & Sons, USA
R2	Computer Aided Design Laboratory by M. N. Sessa Praksh& Dr. G. S. Servesh – Laxmi Publications.
R3	Engineering Graphics by P. J. Sha – S. Chand & Co.
R4	Introduction to AutoCAD 2018 for Civil Engineering Applications by Nighat Yasmin, SDC Publications

Online Resources:

Sl.No	Website Link
1	https://www.autodesk.in/campaigns/autocad-tutorials
2	https://thesourcecad.com/autocad-tutorials/
3	https://all3dp.com/1/autocad-tutorial-for-beginners/
4	https://civilengineerspk.com/autocad-video-tutorials/